

Jayashri Viswanathan

+1 (510) 402-7134 | viswanaj@gmail.com | <https://www.linkedin.com/in/viswanaj/>

EDUCATION

Stanford University Master of Science, Neurosciences	2025
Cold Spring Harbor Laboratory Certificate, Ion Channels	2023
University of California, Davis Bachelor of Science, Biochemistry and Molecular Biology, <i>Summa Cum Laude</i>	2021

RESEARCH EXPERIENCE

Dr. Ivan Soltesz Lab Stanford University, Department of Neurosurgery - Identified a rapid synaptic adaptation mechanism relevant for learning / memory disorders. - Generated novel hypotheses to resolve apparent inconsistencies regarding endocannabinoid-dependent signaling. - Developed python package for automated signal detection and classification, reducing analysis time by 90%. - Optimized experimental throughput, completing 45+ long-term electrophysiological recordings in 2 weeks.	2024 – 2025
Dr. John R. Huguenard Lab Stanford University, Department of Neurology - Established and validated high-precision patch-clamp assays on brain slices to study inhibitory signaling. - Designed and implemented genetically encoded biosensors to visualize neuromodulator responses in real-time. - Built and integrated MATLAB software for image processing and signal detection in 1 month. - Led an intensive two-week intensive training program in electrophysiology for 15 graduate students. - Discovered regional cAMP oscillations in the thalamus, forging avenues for understanding sleep and seizure. - Delivered invited talks at multiple seminars and international conferences.	2022 – 2024
Dr. David E. Olson Lab University of California Davis, Department of Chemistry - Collaborated cross-functionally with chemists and behavioral neuroscientists in a drug discovery context. - Validated a novel fluorescent probe to visualize LSD target-receptor interactions in live cells. - Characterized an intracellular receptor assembly that may affect drug efficacy via co-immunoprecipitation. - Discovered that an opioid administered systemically potentiates behaviors elicited by a serotonergic agonist. - Constructed dose response curves and aided behavioral characterization of 25+ unique compounds. - Delivered presentations of research on 6+ instances.	2018 – 2021

RESEARCH TECHNIQUES

Data Analysis: Python, R, MATLAB, Image J, JMP, Prism / Graphpad, Adobe Illustrator, Microsoft Office, Apple iWork, Anaconda Navigator, Clampex / Clampfit

Molecular Biology: PCR, ddPCR, SDS-PAGE, co-IP, Western blotting, DNA extraction, RNA extraction, tissue culture (primary neuronal culture, HEK293), electroporation, transfection, high-throughput assay development (neuronal morphology, BRET / FRET, MTT), immunocytochemistry, UV-Vis spectroscopy

Neuroscience: Patch-clamp electrophysiology, stereotactic surgery, fiber photometry, optogenetics, EEG, signal detection, rodent behavior (forced swim test, open field, head-twitch response), immunohistochemistry

Microscopy: fluorescence, confocal, high-content screening (CX7), DIC

HONORS AND AWARDS

Sigma Xi	2026
Koum Foundation Fellowship	2025
Bertarelli Foundation Fellowship	2023
Molecular and Cellular Biology Departmental Citation. UC Davis	2021
College of Biological Sciences Distinguished Scholar Award, UC Davis	2021
Chancellor’s Award for Excellence in Undergraduate Research	2021
UC Berkeley Graduate Fellowship	2021
UC Davis Endorsee, Rhodes Scholarship	2020
UC Davis Endorsee, Marshall Scholarship	2020
Hanson Family Undergraduate Research Publication Award	2020
Phi Beta Kappa	2020
UC Davis Provost’s Undergraduate Fellowship	2020
Barry M. Goldwater Scholarship	2020
UC Davis Provost’s Undergraduate Fellowship	2019
Best Presentation in Chemical Biology, Richard LaRock Conference	2019
Royal Society of Chemistry Best Chemical Science Poster, Miller Symposium	2019
Barabara Chapman Best Poster Presentation, 2 nd Place, UC Davis Neurofest	2019
Dean’s List, College of Biological Sciences	2019

PEER-REVIEWED PUBLICATIONS

Aarrestad IK, Cameron LP, Fenton EM, Casey AB, Rijsketic DR, Patel SD, Sambyal R, Johnson SB, Ly C, **Viswanathan J**, Barragan EV, Lozano SA, Seban N, Hu H, Powell NA, Chytil M, Meyer R, Rose D, Hempel C, Olson E, Hansen HD, Madsen CA, Knudsen GM, Redd C, Wheeler DG, Guanzon N, Muir J, Hennessey JJ, Quon G, McCorvy JD, Gandhi SP, Rasmussen K, Liston C, Gray JA, Heifets BD, Nord AS, Kim CK, Olson DE.
Compound-induced neuroplasticity without immediate early gene activation. *Nature Neuroscience*, 2025

Cameron LP, Tombari RJ, Lu J, Pell AJ, Hurley ZQ, Ehinger Y, Vargas MV, McCarroll MN, Taylor JC, Myers-Turnbull D, Liu T, Yaghoobi B, Laskowski LJ, Anderson EI, Zhang G, **Viswanathan J**, Brown BM, Tjia M, Dunlap LE, Rabow ZT, Fiehn O, Wulff H, McCorvy JD, Lein PJ, Kokel D, Ron D, Peters J, Zuo Y, Olson DE.
A non-hallucinogenic psychedelic analogue with therapeutic potential. *Nature*, 2021

Dunlap LE, Azinfar A, Ly C, Cameron LP, **Viswanathan J**, Tombari RJ, Myers-Turnbull D, Taylor JC, Grodzki AC, Lein PJ, Kokel D, Olson DE.
Identification of Psychoplastogenic N,N-Dimethylaminoisotryptamine (isoDMT) Analogues through Structure–Activity Relationship Studies. *J. Medicinal Chemistry*, 2020

Lau V, Chiang V, Kardani O, Kim A, Kwan E, Panchal A, **Viswanathan J**, Afghani B.
Efficacy of Steroids in Preventing Long-Term Renal Involvement in Pediatric Patients with Henoch Schonlein Purpura (HSP): A Comprehensive Literature Review. *J. Investigative Medicine*, 2017

TEACHING EXPERIENCE

Teaching Assistant, Stanford Immersive Neuroscience Bootcamp	2022
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CONFERENCE PRESENTATIONS

cAMP Oscillates in Thalamic Slice Preparations. Gordon Research Conference, Thalamocortical Interactions, 2024.
Author: **Viswanathan J**

cAMP Oscillates in Thalamic Slice Preparations. Gordon Research Seminar, Thalamocortical Interactions, 2024.
Author: **Viswanathan J**

Developmental progression of prefrontal circuit dysfunction in a SHANK3^{-/-} mouse model of Autism Spectrum Disorder. BAAC Annual Research Symposium, 2022.
 Authors: Devienne G, Vantomme G, **Viswanathan J**, Huguenard J

Developmental progression of prefrontal circuit dysfunction in a SHANK3^{-/-} mouse model of Autism Spectrum Disorder. Society for Neuroscience, 2022.
 Authors: Devienne G, Vantomme G, **Viswanathan J**, Huguenard J

Developmental progression of prefrontal circuit dysfunction in a SHANK3^{-/-} mouse model of Autism Spectrum Disorder. Gordon Frontal Cortex, 2022.
 Authors: Devienne G, Vantomme G, **Viswanathan J**, Huguenard J

Integration of dopamine and serotonin signals in the Nucleus Accumbens gates learning. Society for Neuroscience, 2022.
 Authors: Pinto DFC, Pomrenze MB, Guo MY, **Viswanathan J**, Bentzley BS, Eshel N, Malenka RC

Simultaneous genetic access to the brain's dopamine and serotonin systems reveals neuromodulatory dynamics underlying Pavlovian learning. Dopamine, 2022.
 Authors: Pinto DFC, Pomrenze MB, **Viswanathan J**, Guo MY, Bentzley BS, Eshel N, Malenka RC

Simultaneous genetic access to the brain's dopamine and serotonin systems reveals neuromodulatory dynamics underlying Pavlovian learning. Gordon Optogenetics, 2022.
 Authors: Pinto DFC, Pomrenze MB, **Viswanathan J**, Bentzley BS, Eshel N, Malenka RC

Psychedelics as Potential Therapeutics for Alzheimer's Disease. UC Davis Annual Alzheimer's Disease Symposium, 2019.
 Authors: **Viswanathan J**, Olson DE

In Vivo and In Vitro Evaluation of Synthetic Iboga Analogs for Treatment of Neuropsychiatric Disorders. Richard LaRock Undergraduate Research Conference, 2019.
 Authors: **Viswanathan J**, Cameron LP, Olson DE

Neural Plasticity-Promoting Analogs of Iboga Alkaloids Identified Through Function-Oriented Synthesis. Undergraduate Research Conference, 2019.
 Authors: Pell AJ, **Viswanathan J**, Cameron LP, Olson DE

Psychedelics for Treating Depression and Anxiety. UC Davis Neurofest, 2019.
 Authors: **Viswanathan J**, Cameron LP, Olson DE

Psychedelics for Treating Depression and Anxiety. UC Davis Decision Day, 2019. A
 uthors: **Viswanathan J**, Cameron LP, Olson DE

Psychedelics for Treating Neuropsychiatric Disorders. R. Bryan Miller Symposium, 2019.
 Authors: **Viswanathan J**, Cameron LP, Olson DE

PROFESSIONAL SERVICE

Seminarian

Stanford University, Department of Molecular and Cellular Physiology

2024

Student Retreat Organizer

Stanford University, Neurosciences Interdepartmental Program Retreat

2022